Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

# SDC Hackathon

# Technical Tools Declaration

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Submission Deadline: 25 October 2026**

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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**SDC HACKATHON 2026**

# Technical & Tools Declaration

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## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | Pentagrammers |
| **Project / Solution Name** | Orbyte Smart Reorder Assistant |
| **Name (Leader)** | CHA ZHI YI |
| **Email & Phone (Leader)** | chazyee0306@gmail.com 0169236287 |
| **University Affiliation(s)** | Universiti Teknikal Malaysia Melaka (UTeM) |
| **Live Prototype URL** | https://e18001d8-ab3c-4883-9464-81086cc8e9f0-00-95b61jke4hrr.sisko.replit.dev/ |
| **Code Repository URL (public / access granted to judges)** | https://github.com/Zhiyee-63/orbyte-smart-inventory |

## 2. Architecture & Technology Stack

### 2.1 System Architecture Overview

*Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.*

***[Maximum 200 words]***

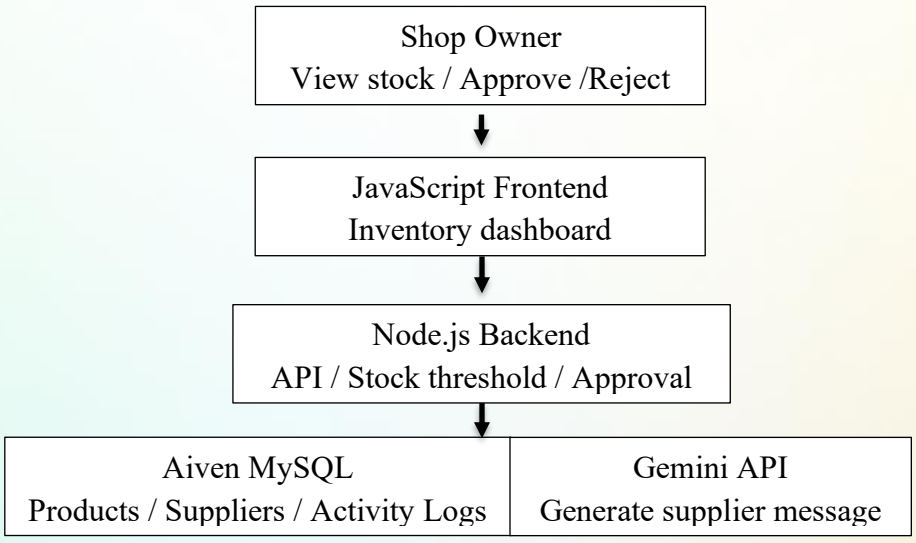
The Orbyte Smart Reorder Assistant consists of a JavaScript frontend, a Node.js backend, an Aiven-hosted MySQL database, the Google Gemini API, and WhatsApp Click-to-Chat (`wa.me`).

The shop owner uses the frontend dashboard to view inventory and manage replenishment requests. When a sale is recorded, the backend updates the product quantity in MySQL and checks whether stock has fallen below the predefined minimum level.

If low stock is detected, the backend retrieves relevant product and supplier information and sends the context to the Gemini API to generate a WhatsApp restocking message. The proposed message appears on the dashboard for the owner to review, edit, approve, or reject.

After approval, the system opens the supplier's WhatsApp chat through a `wa.me` link with the message pre-filled. The owner must manually send the message through WhatsApp. Relevant actions and outcomes are recorded in the database as activity logs.

The application is planned to run on Replit, with API credentials and database connection details stored securely in environment variables.



### 2.2 Languages, Frameworks & Libraries

*List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.*

| Name & Version | Purpose in Project | License |
|---|---|---|
| JavaScript (ECMAScript) | Programming language for frontend and backend development | ECMAScript specification |
| React 19.3.1 | Builds the frontend inventory dashboard and user interface | MIT License |
| Vite 8.3.4 | Frontend development server and build tool | MIT License |
| Node.js 24.13.0 | Runs the backend server and API endpoints | MIT License |
| Express 5.3.0 | Handles backend HTTP requests and API routes | MIT License |
| cors 2.8.6 | Allows the frontend to communicate with the backend API | MIT License |
| mysql2 3.24.6 | Connects the Node.js backend to the MySQL database | MIT License |
| MySQL Community Server 8.4.11 | Stores products, suppliers, restock orders, and activity logs | GPL v2 |
| Aiven for MySQL | Provides cloud hosting for the MySQL database | Aiven Terms of Service |
| MySQL Workbench 26.10.0 | Creates database tables, loads demo data, and checks records | GPL v2 (Community Edition) |
| Google Gemini API (gemini-3.1-flash-lite, gemini-3.5-flash, gemini-3.8-flash) | Generates contextual WhatsApp replenishment messages in Bahasa Melayu | Google APIs Terms of Service |
| `@google/genai` 2.31.0 | Node.js SDK for calling the Gemini API | Apache 2.0 |
| WhatsApp click-to-chat link (wa.me) (n/a URL feature) | Open the supplier chat with the message pre-filled after approval | Free public feature |

| Name & Version | Purpose in Project | License |
| --- | --- | --- |
| Replit | Development, hosting, and deployment of the application | Replit Terms of Service |

## 3. APIs & Cloud Services

### 3.1 External APIs

*List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.*

| API & Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
|---|---|---|
| Google Gemini API | Generate contextual WhatsApp replenishment messages when stock is low | Free tier |
| WhatsApp click-to-chat link (wa.me) | After the owner approves, opens the supplier's WhatsApp chat with the message pre-filled, and the owner taps Send | Free |
| Aiven for MySQL | Provides managed MySQL database connectivity | Free trial |
| Replit | Hosts and deploys the Node.js application | Core plan (promotional code; free for the team) |

### 3.2 Cloud Platforms & Deployment

*List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed*

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| Replit | Development, hosting, and deployment of the Node.js backend and React frontend | Core plan (free through promotional code); region not specific |
| Aiven for MySQL | Cloud hosting for the MySQL database storing products, suppliers, restock orders, and activity logs | Free trial; Region: (Bengaluru, India) |
| Google Gemini API | Generates contextual WhatsApp replenishment messages | Free Plan; Mumbai, India |
| WhatsApp click-to-chat link (wa.me) | After the owner approves, opens the supplier's WhatsApp chat with the message pre-filled, and the owner taps Send | No account or API key |

## 4. AI Tools, Models & Datasets

### 4.1 AI Models Used in the Solution

*List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.*

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
| Gemini Model (gemini-3.1-flash-lite, gemini-3.5-flash, gemini-3.8-flash) | Google Gemini API | Pre-trained; generates WhatsApp restocking messages based on inventory information | Google Gemini API Terms of Service |

### 4.2 Datasets

*List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.*

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| Synthetic Product Inventory Data (22 products, `Products` table) | Created by the team: `backend/database/seed.sql` in the GitHub repository | Synthetic | Team-created; no external dataset license. (e.g. Milo, Gardenia, Lifebuoy) are used only as realistic labels; barcodes and stock numbers are made up. | N |
| Synthetic Sales Records (14 days, 308 records, `Activity_Logs` table) | Created by the team: generated by `backend/database/seed.sql` | Synthetic | Team-created; no external dataset license. | N |

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| Synthetic Supplier Data (5 suppliers, `Suppliers` table) | Created by the team: `backend/database/seed.sql` | Synthetic | Team-created; no external dataset license. Supplier names, contact names and phone numbers are fictional. | Y |
| Synthetic Reorder Data (`Restock_Orders` table) | Created by the team: `backend/database/seed.sql` | Synthetic | Team-created; no external dataset license | N |
| Synthetic Activity Log Data (`Activity_Logs` table) | Created by the team: `backend/database/seed.sql` | Synthetic | Team-created; no external dataset license | N |

### 4.3 AI-Assisted Development Tools

*Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.*

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

| **Tool & Version** | **Used For** | **Extent of Use (minor / moderate / major)** |
|---|---|---|
| ChatGPT GPT-6 | Helped draft project documentation, explain technical concepts, and improve written content | Moderate |
| Claude Sonnet 5.5 | Assisted with backend development, including code generation, debugging, and solving backend programming problems | Moderate |
| Google Gemini 3.1 Pro | Assisted with drafting the frontend UI | Moderate |
| Google Gemini 3.1 Pro | Helped implement server-side functionality and debug backend errors. | Moderate |
| Gemini 4 Argon | Fixing connection to backend problem | Moderate |
| Claude Opus 5.5 | Assisted with database design, database queries, and solving database-related programming problems | Moderate |
| Google Gemini 3.6 Flash | Assisted in implementing backend logic and resolving technical issues. | Moderate |

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

***[Maximum 100 words]***

The team reviewed AI-generated code and content to ensure they matched the project requirements. AI-assisted code was checked and tested during development, and errors were corrected when identified. The team reviewed AI-generated database queries, backend logic, and frontend UI suggestions before using them in the application. AI-generated restocking messages are intended to be reviewed by the shop owner before sending. The team remains responsible for the accuracy, functionality, and quality of the final project rather than relying entirely on AI-generated output.

## 5. Sponsor-Provided Resources

*Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.*

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

| Sponsor | Resource (API / credits / platform / data) | How It Was Used in the Project |
|---|---|---|
| Replit | Replit Agent (AI credits: $20.00) and cloud platform | Used 1.7% of credits ($0.34 of $20.00), all on Replit Agent. It helped set up the Node.js Express backend, connect the MySQL database, and add the Google Gemini API endpoint. The backend was also run and tested on Replit. |

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

***[Maximum 100 words]***

Replit Agent helped accelerate backend development by setting up the Node.js Express backend, connecting the MySQL database, and integrating the Google Gemini API endpoint. Replit also provided a cloud environment for running and testing the application. Although these resources were useful for development, they were not essential to the core functionality of the prototype. The backend could be developed using other coding tools, while the application could be deployed on alternative cloud platforms. However, replacing Replit would require additional setup and configuration.

## 6. Security, Privacy & Responsible Use

### 6.1 Data Handling & Privacy

*What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.*

***[Maximum 150 words]***

The application uses synthetic product inventory, sales, and supplier data for demonstration and testing. It does not require real customer personal data. Inventory information is stored in a MySQL database hosted on Aiven. Relevant inventory and supplier information may be sent to the Google Gemini API to generate restocking message drafts. After approval, WhatsApp Click-to-Chat (`wa.me`) opens the supplier's WhatsApp chat with a pre-filled message, which the shop owner sends manually. API keys and database credentials should be stored securely as environment variables and must not be exposed in source code. Access to the database and third-party services should be restricted to authorized team members. Data retention depends on database configuration and third-party service terms. In a real deployment, the application should comply with Malaysia's Personal Data Protection Act (PDPA), where applicable, and avoid sending unnecessary personal or sensitive information to external services.

### 6.2 Known Limitations & Risks

*Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.*

***[Maximum 150 words]***

The application relies on the Google Gemini API to generate restocking message drafts. The AI may produce inaccurate information, incorrect product quantities, or unsuitable messages. API usage may also be affected by rate limits, service outages, or changes in provider availability. Since the application depends on a single AI provider, switching providers may require code modifications. To reduce these risks, the system uses inventory data stored in the database as the basis for generating messages. Shop owners can review, edit, approve, or reject AI-generated messages before they are sent. This human approval step helps prevent incorrect or unintended messages from being sent automatically. The system should also validate inventory quantities and handle API errors gracefully. As this is a prototype, its accuracy, reliability, and performance require further testing before real-world deployment.

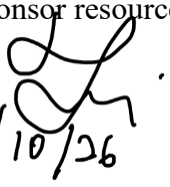
## 7. Compliance & Readiness Checklist

*Tick all boxes that apply before submitting.*

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

## 8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] 

[Date] 25/10/26